

Milestone 2: Student Portal & Personalized Learning Ecosystem

1. State Management & Session Context Lifecycle

- **Global Context Provider:** Programmed AuthContext.jsx distributing active student profile structures, loading statuses, and token records down the React node hierarchy.
- **Session Token Storage:** Enabled session tokens to be saved inside localStorage or session cookies, keeping students logged in on page reloads. Built clear() hooks for sign-outs.
- **Axios Network Interceptors:** Set up api.js wrapping outgoing requests. Configured interceptors appending the bearer authentication token: `config.headers['Authorization'] = `Bearer ${token}``.

2. Implemented Features: Student Dashboards & Custom Profiles

- **Personalized Student Dashboard:** Developed StudentDashboard.jsx displaying metrics on course enrollments, current quest progress, study schedules, and recommended pathways.
- **Student Profile Management:** Programmed profile editing views. Students can upload profiles, add professional summaries, list accomplishments, and link external resumes.
- **Skill Analytics Matrix:** Created visual skill matrix dashboards displaying user competency scores across various domains based on completed assessments.

3. Implemented Features: Core Learning Hub & Course Management

- **Course Catalog Directory:** Implemented CourseList.jsx displaying course catalogs. Supported filter attributes like category, difficulty level, and keywords.
- **Learning Hub & Paths:** Mapped dynamic courses with syllabus maps and navigation tracks. Includes widgets to read class notes and view video training lectures.
- **Course Enrollment Manager:** Set up CourseController endpoints tracking course enrollment records, initializing progress tracking parameters to zero upon registration.

4. Implemented Features: Micro-Learning & Gamification

- **Daily Quests System:** Engineered quest parameters verifying logins, flashcard reviews, and sandbox executions. Updates student score profiles.
- **Flashcards Module:** Created interactive quiz panels allowing students to flip flashcard blocks, check key concepts, and rate their understanding level.
- **Sandbox Playgrounds:** Built a playground layout executing scripts for fast testing without impacting project files.

5. Implemented Features: Assessments & Certification Tracking

- **End-of-Unit Assessments:** Programmed quiz controllers delivering multiple-choice tests, calculating pass rates, and tracking certification eligibilities.
- **Certificate Tracking System:** Created Certificate.java entity mapping earned certifications to student accounts, displaying credentials with print options.

6. Deliverables & Technical Verification

- **[X]** Global session state wrapper fully surrounding portal routing.
- **[X]** Responsive layout sidebar, navbar, and card components deployed.
- **[X]** JPA repository interfaces CourseRepository and EnrollmentRepository active.
- **[X]** REST controllers for course browsing, enrollment, and profile updates verified.